WashU Medicine

ENHANCE CVH

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On behalf of the ENHANCE CVH Investigators

ENHANCE-CVH

NYU

parents as teachers

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Disclosures

Grants: R01HL168771, R01HL144708, D43TW011976, UH3HL152381, K24HL175228, Resolve to Save Lives, R33HL139852, 24FIM1268045, 26FIM1578918

Intellectual: Pending patents for heart failure polypills

Travel: World Heart Federation

Institutional: George Health Enterprises Pty Ltd (GH) and its subsidiary, George Medicines Pty Ltd, have received investment funds to develop fixed-dose combination products, including combinations of blood pressure-lowering drugs. GH is the social enterprise arm of The George Institute for Global Health where I have a secondary appointment.

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Program overview

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Nigeria has the highest burden of maternal morbidity and mortality.

- Nigeria: 528 (95% UI: 351, 815) maternal deaths per 100,000 lives birth in 2019
- Global average: 152 per 100,000 in 2019
- SDG Target 3.1: <70 per 100,000 live births by 2030

1 out 3 of maternal deaths in Nigeria are due to hypertension.

Thus, we adapted the Enhancing Intergenerational HeAlth in Nigeria: Peripartum as Critical Life StagE for CardioVascular Health (**ENHANCE-CVH**) cluster randomized (1:1) trial.

GBD 2019. *Lancet*. 2020;396:1160-1203.
Tukur J, et al. *Eclinical Med*. 2022;47:101411.

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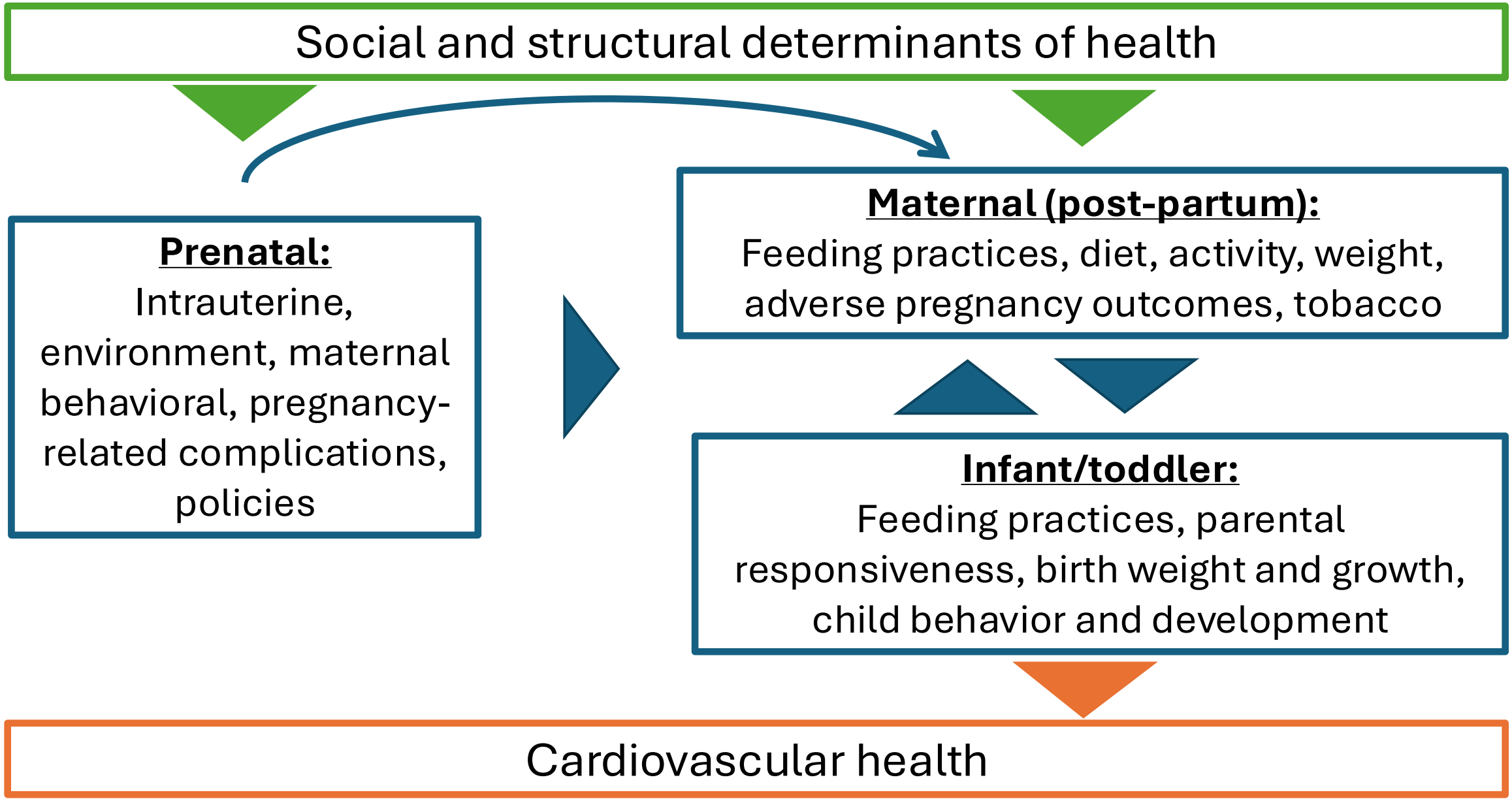

Conceptual framework for intergenerational determinants on cardiovascular health.

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Trial question

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Participants	Pregnant patients up to 20 weeks' gestation and body mass index $\geq$ 26 kg/m^2 recruited from eligible primary healthcare centers in Federal Capital Territory, Nigeria
Intervention	Home visiting based on PAT + HEALTH curriculum
Comparator	Usual care
Key outcomes	Between group difference in change in body weight (1°), maternal CVH (2°) at 18 months, implementation using RE-AIM (1°: reach)

PAT: Parents as Teachers; HEALTH: Healthy Eating Active Living Taught at Home

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Key research findings to date

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Formative research	High acceptability, feasibility, but tailoring, CHEW education and supportive, community awareness, and sustainable funding are needed
Adaptation from ENRICH (using FRAME)	Training content, context for PAT foundational curriculum + HEALTH; CHEWs as training targets
Trial status	n=40 primary healthcare centers randomized in June 2025, 120+/1000 participants recruited

CHEW: community health extension worker; FRAME: Framework for Modification and Adaptations; HEALTH: Healthy Eating Active Living Taught at Home

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Policy impact and scalability

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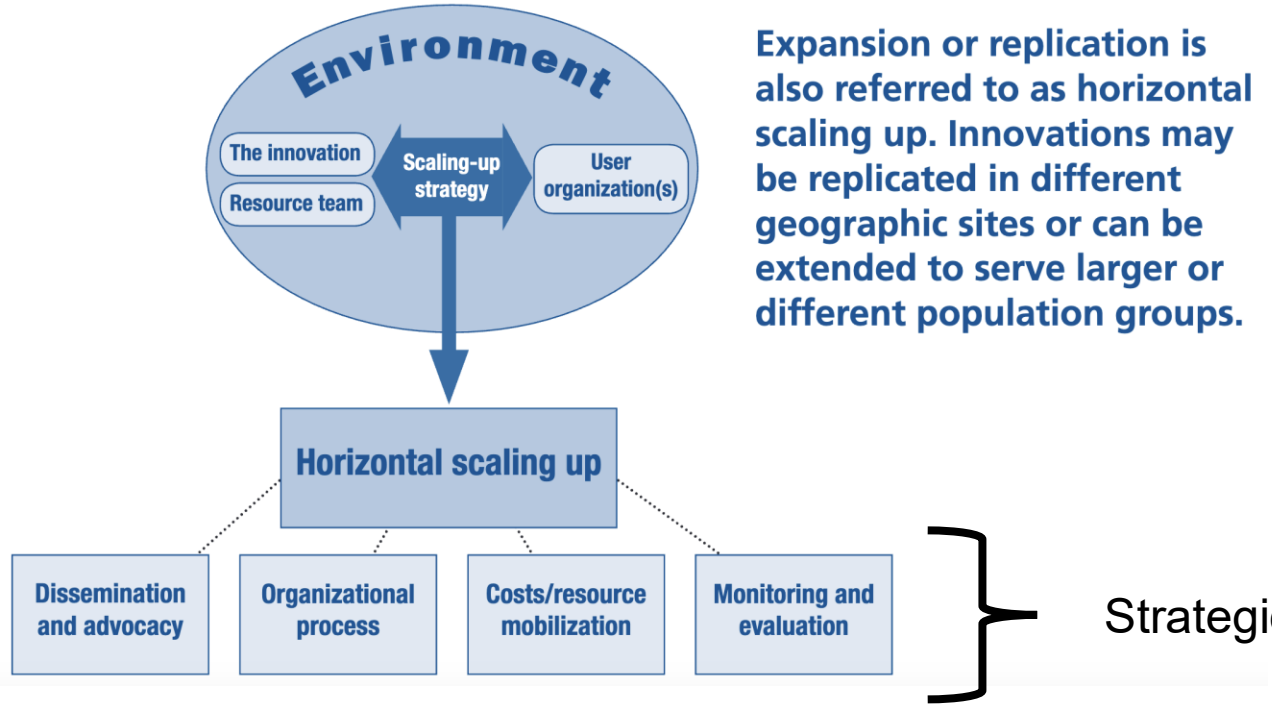
Policy goal: Standardize, sustain, and then scale CHEW-based home visiting if the trial demonstrates effectiveness and implementation.



Expansion or replication is also referred to as horizontal scaling up. Innovations may be replicated in different geographic sites or can be extended to serve larger or different population groups.

Strategic choice areas

WHO ExpandNet framework for horizontal scale-up

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Publications

ESC European Society of Cardiology
European Heart Journal (2025) **46**, 3239–3241
https://doi.org/10.1093/eurheartj/ehaf288
CardioPulse

Global Spotlights

Stakeholder meeting report for ENHANCE CVH: a trial to improve maternal and child cardiovascular health in Nigeria

Cecilia Naa O. Nartey[1], Mark D. Huffman [1,2,*], and Dike B. Ojji [3]; on behalf of the ENHANCE CVH Study Team

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Enhancing Intergenerational HeAlth in Nigeria: Peripartum as Critical Life StagE for CardioVascular Health (ENHANCE-CVH) Study: Findings from Pre-implementation Using the Updated Consolidated Framework for Implementation Research (CFIR 2.0). *Provisional acceptance, Imp Sci Comms*.

Bridging Across Countries : Adapting the PAT HEALTH Curriculum for Improved Maternal and Child Cardiovascular Health in Nigeria. *Submitted*.

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